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(54) **DISPLAY ASSEMBLIES FOR PROVIDING COMPRESSIVE FORCES AT ELECTRONIC DISPLAY LAYERS**

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G06F 1/20 (2006.01)

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(52) **U.S. Cl.**

CPC **G06F 1/181** (2013.01); **G06F 1/20** (2013.01); **G09F 9/33** (2013.01); **H05K 7/20972** (2013.01)

(58) **Field of Classification Search**

CPC G06F 1/181; G06F 1/20; H05K 7/20972

USPC 361/688

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2024/0377054 A1* 11/2024 Gou F21V 29/83

2025/0076558 A1* 3/2025 Schlaupitz G02B 27/0101

* cited by examiner

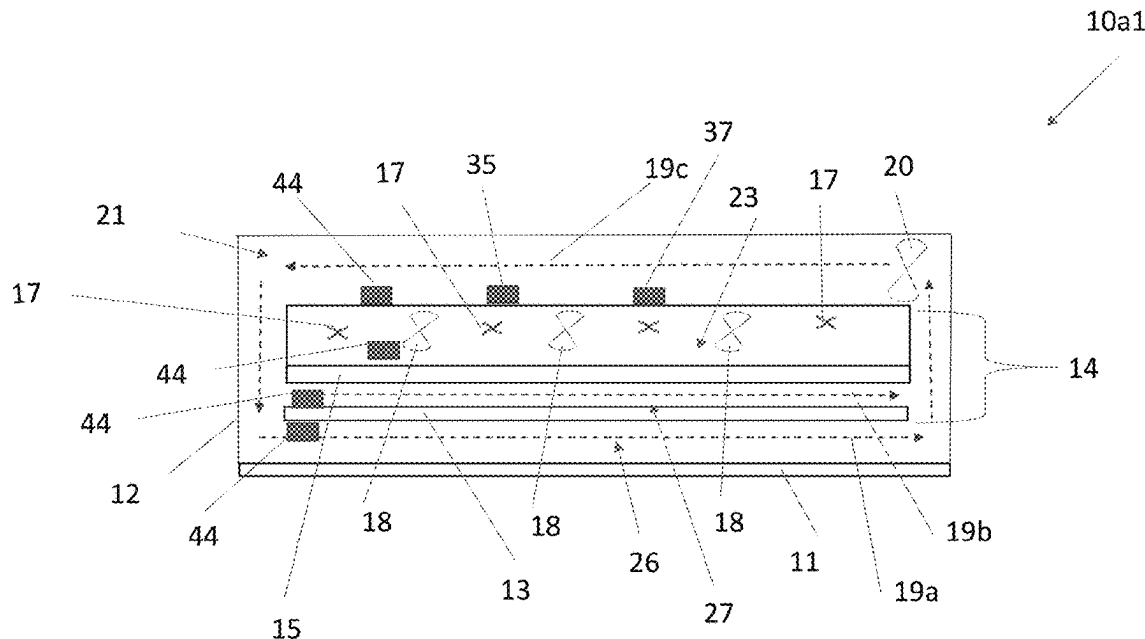
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(57) **ABSTRACT**

Display assemblies for inhibiting delamination of an optical stack are disclosed. A structural framework defines, at least in part, an enclosure for the optical stack, which includes an electronic display layer. One or more airflow passageways are located within the enclosure and define a continuous, interconnected airflow pathway extending along one or more surfaces of the optical stack. One or more fan units are positioned along the continuous, interconnected airflow pathway, and when activated, move gas through the continuous, interconnected airflow pathway and generate compressive forces at the optical stack, thereby reducing the likelihood of delamination.

20 Claims, 12 Drawing Sheets



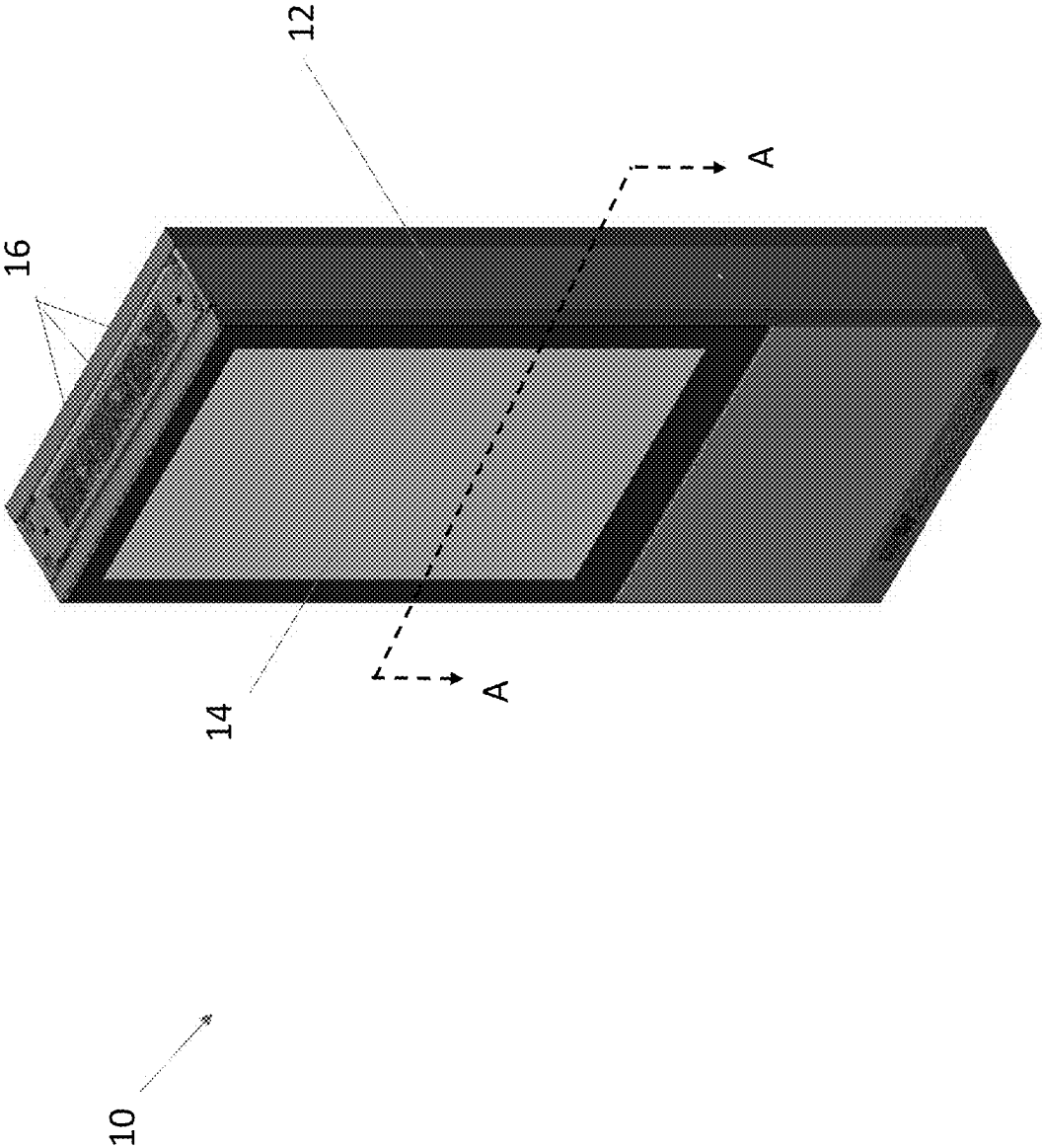


Figure 1

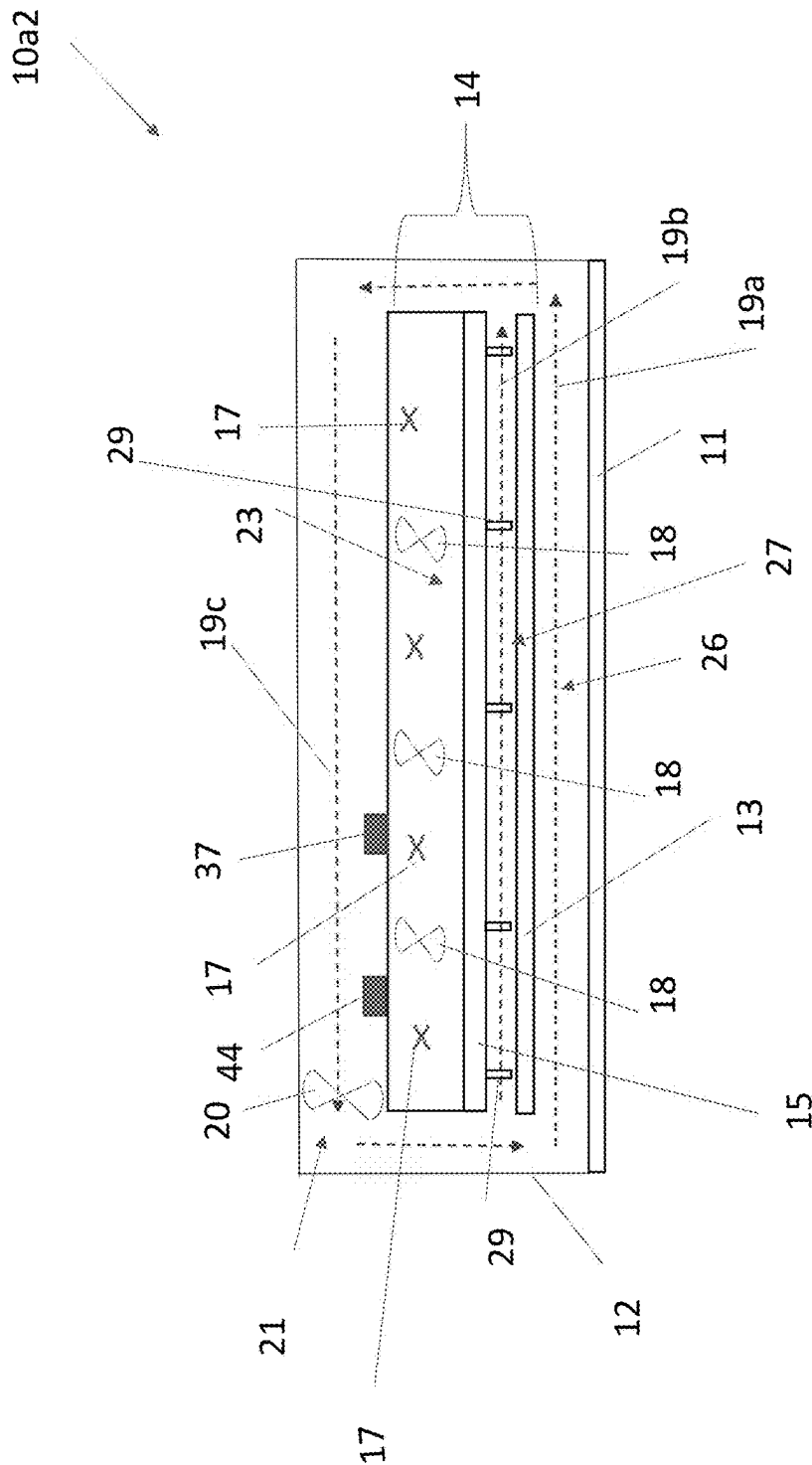


Figure 2B

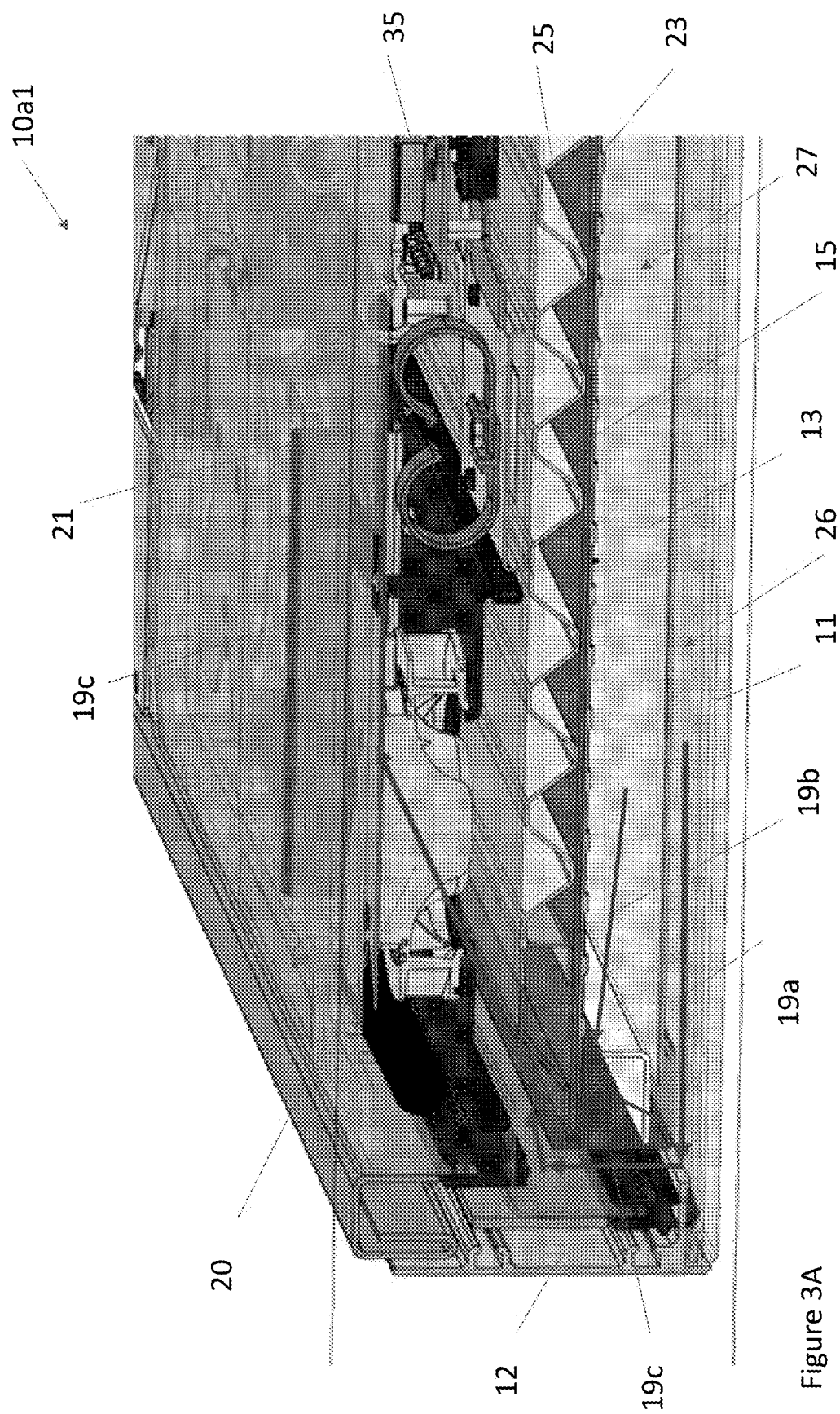


Figure 3A

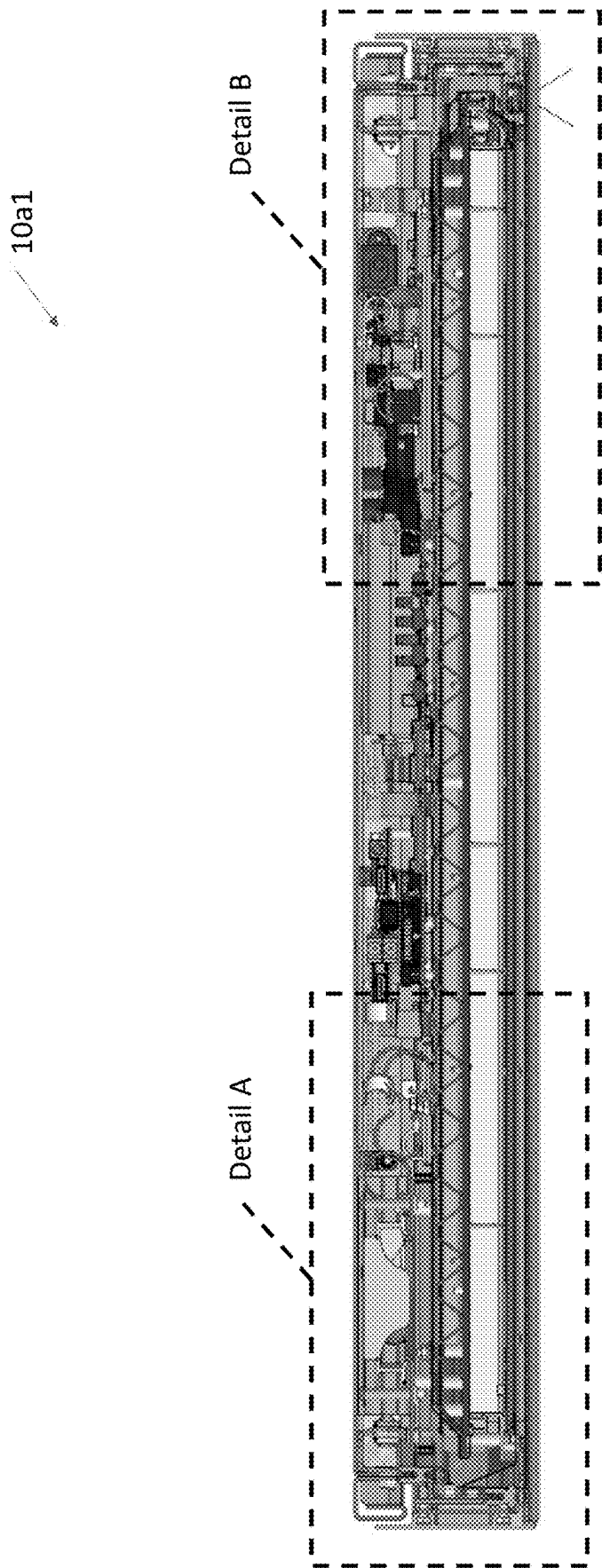


Figure 3A1

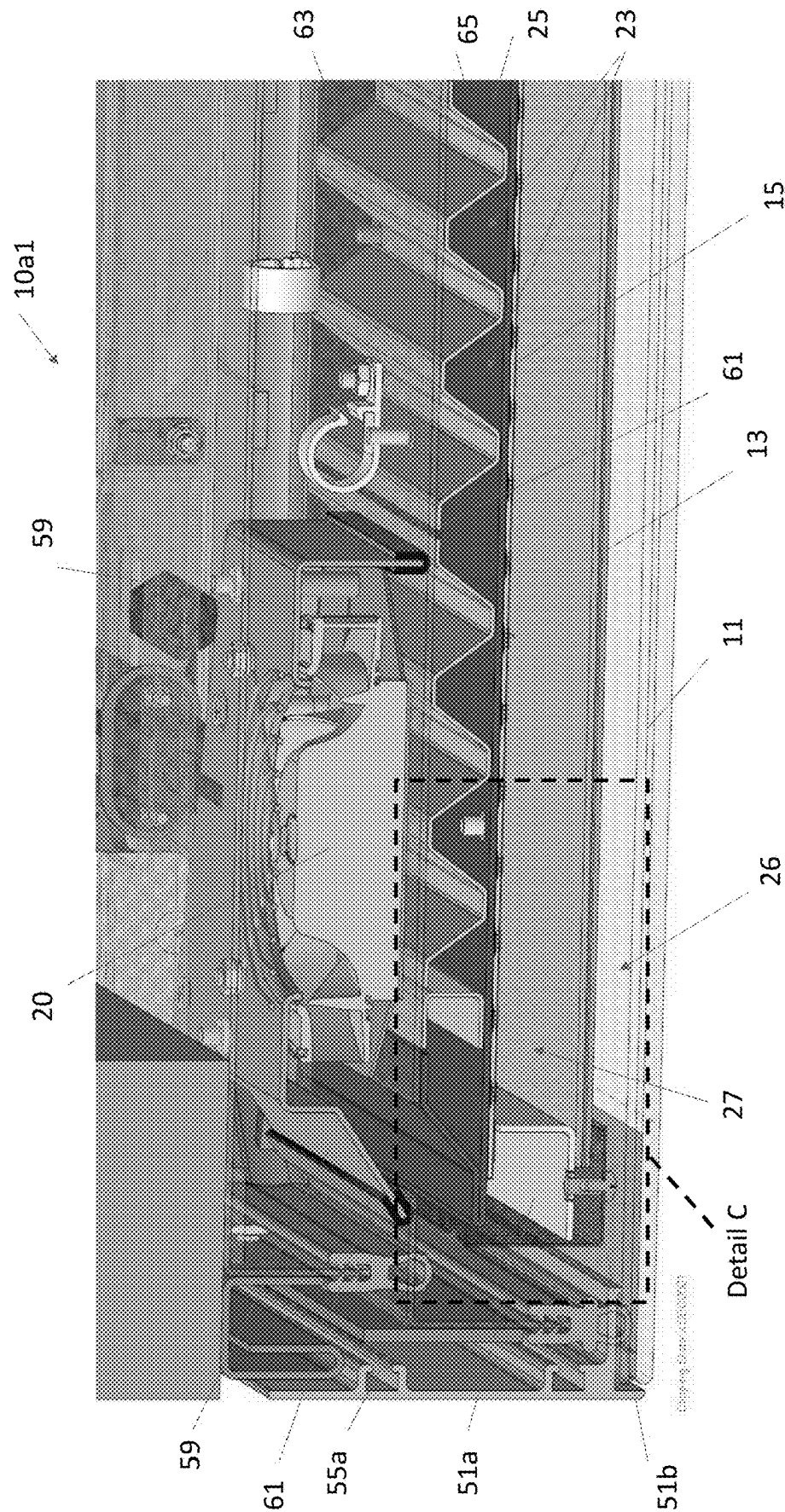


Figure 3A2

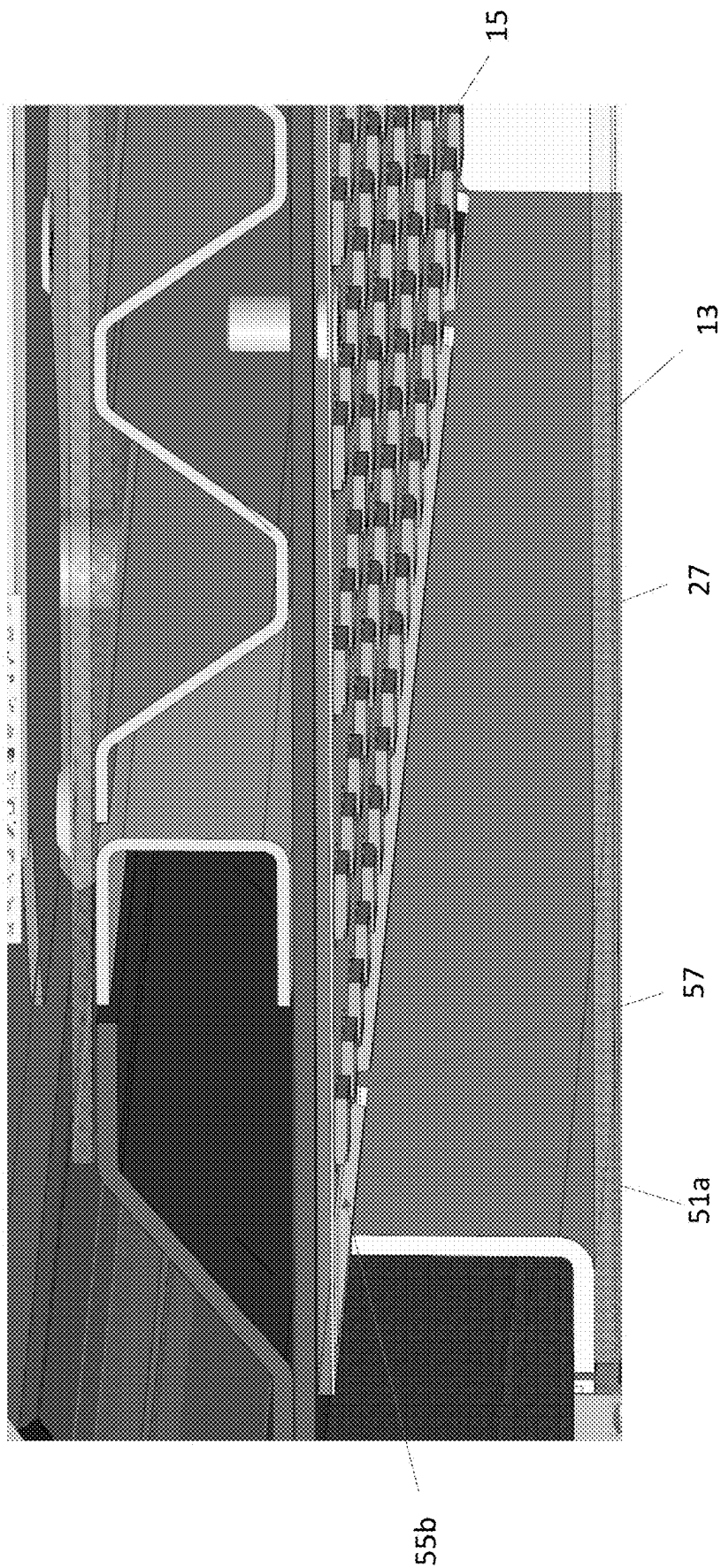


Figure 3A3

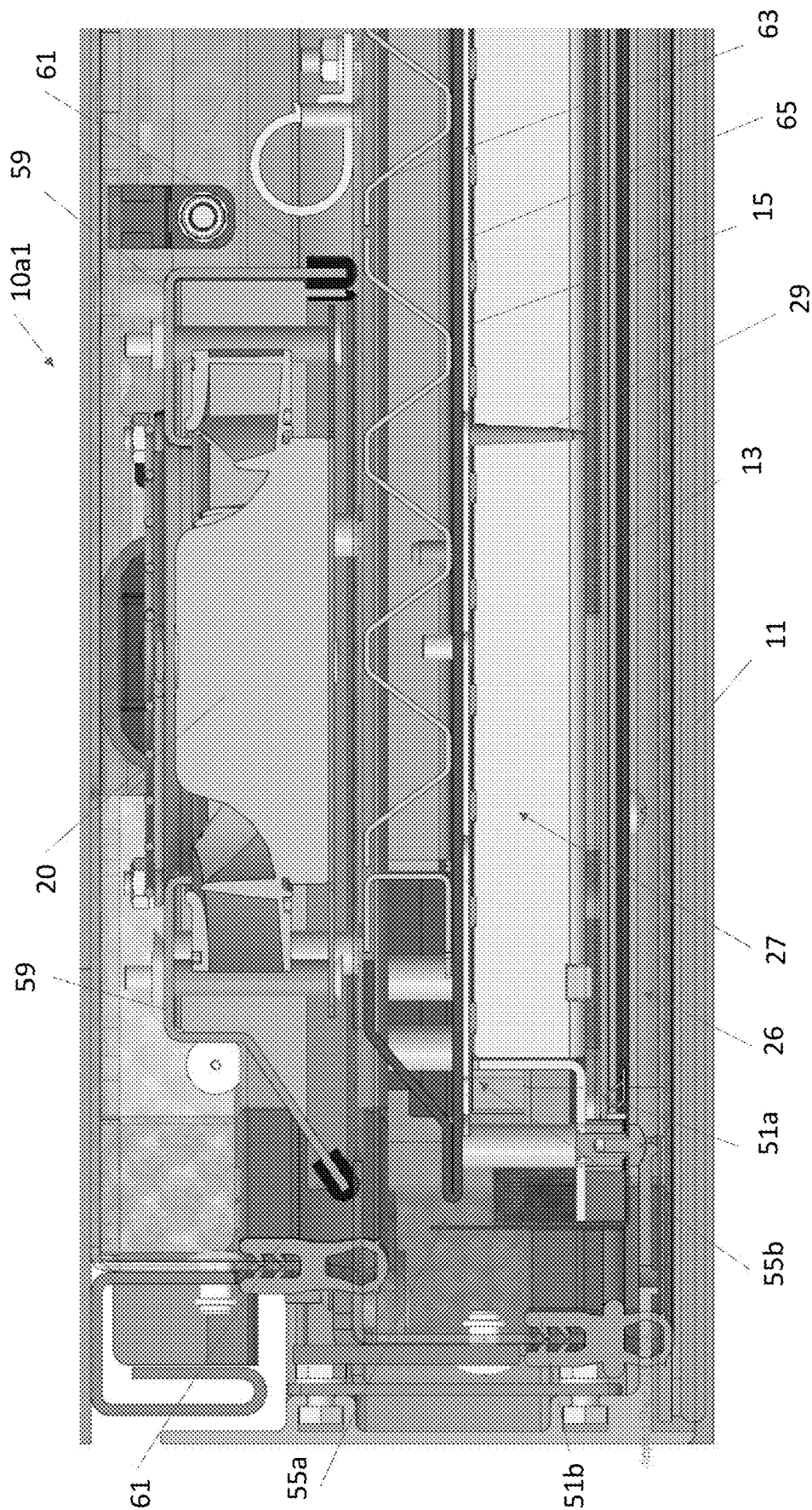


Figure 3A4

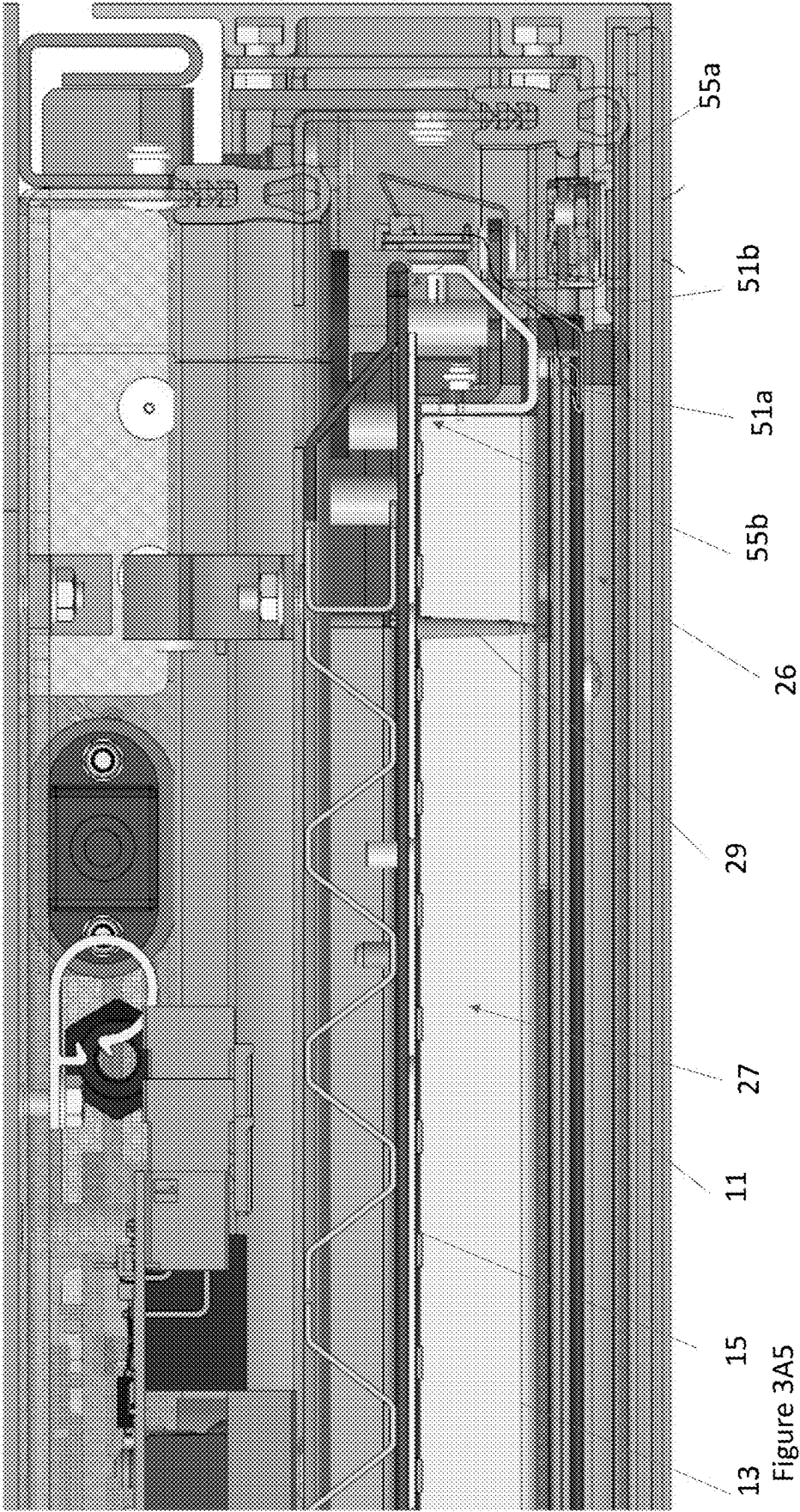


Figure 3A5

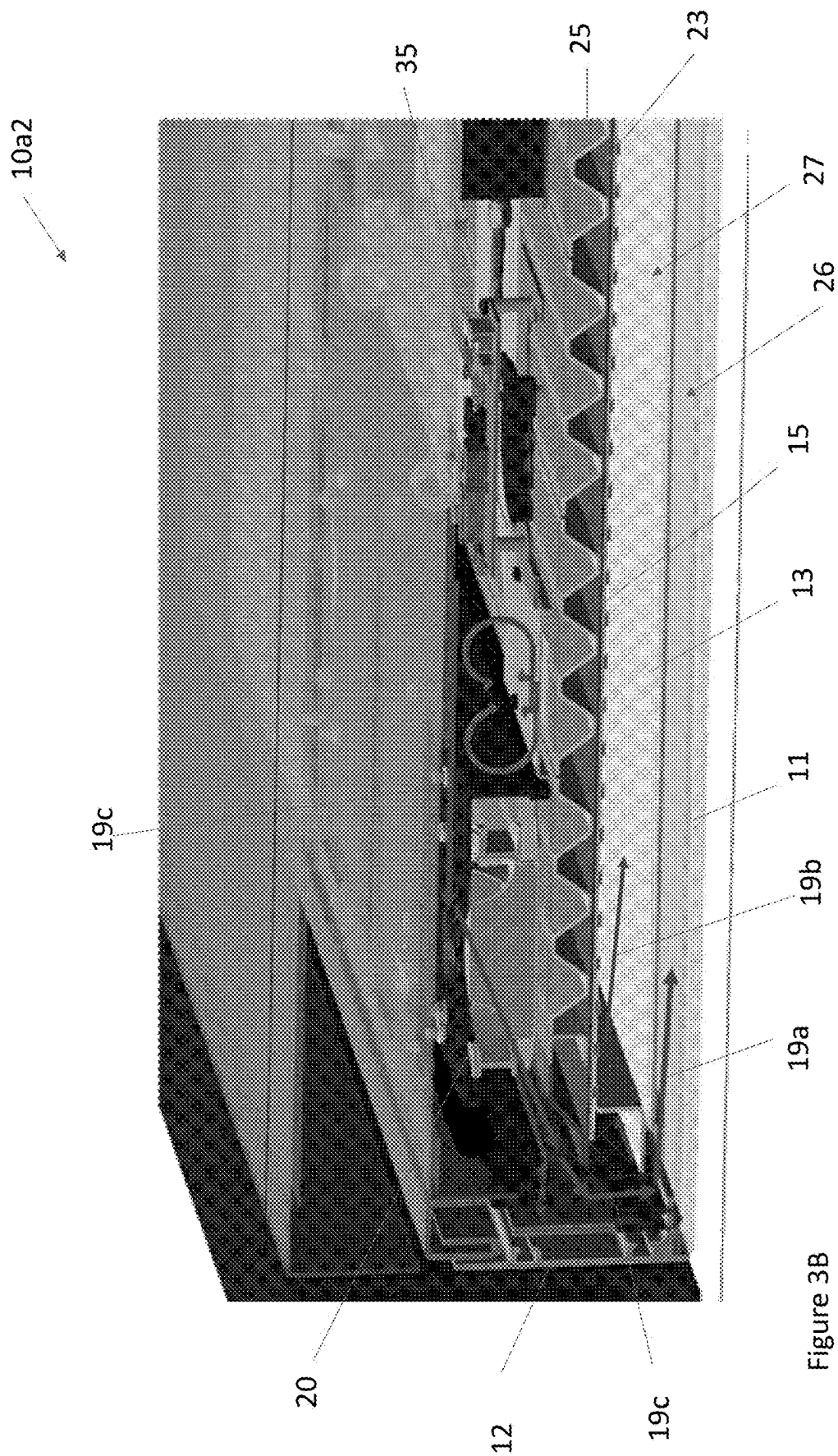


Figure 3B

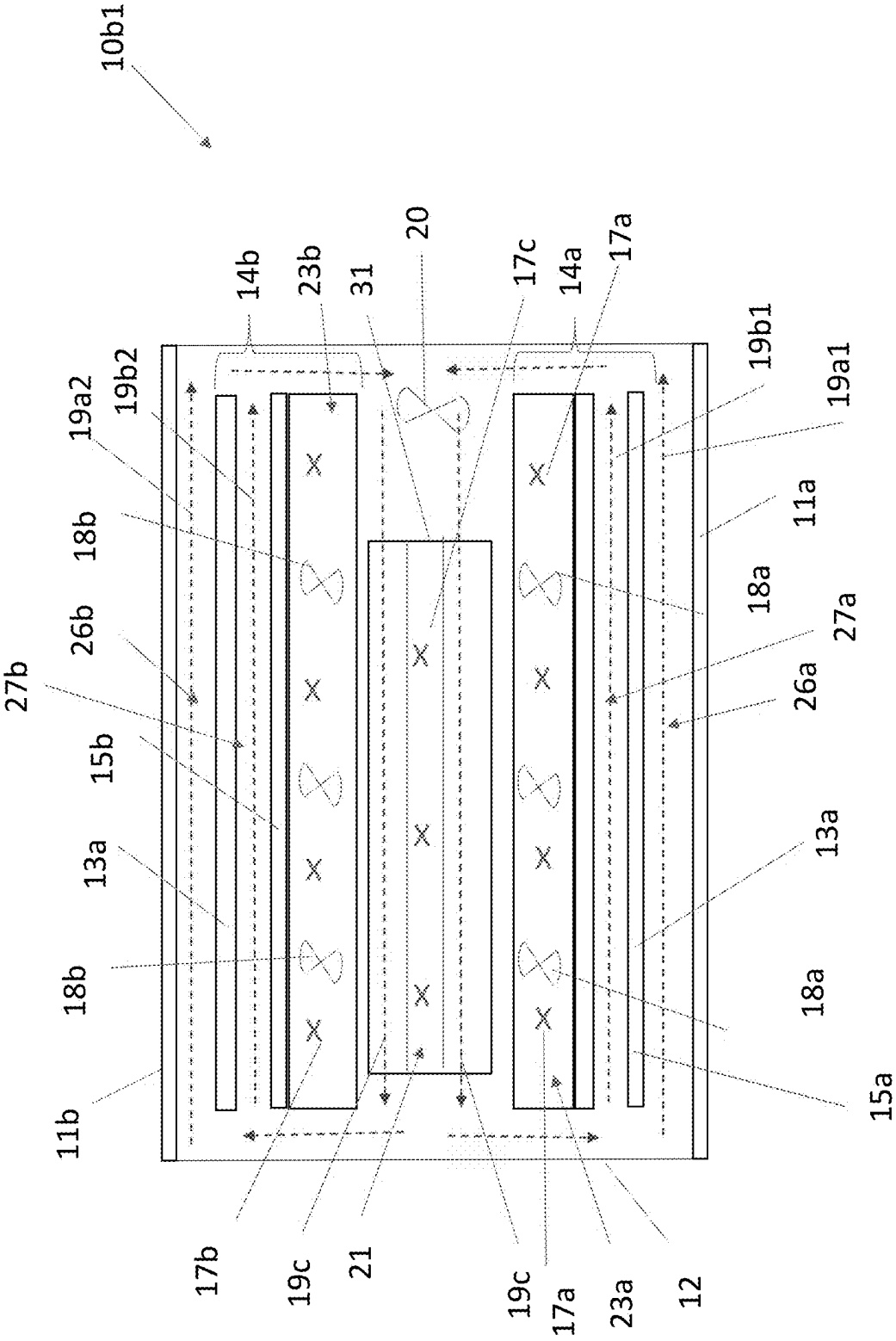


Figure 4A

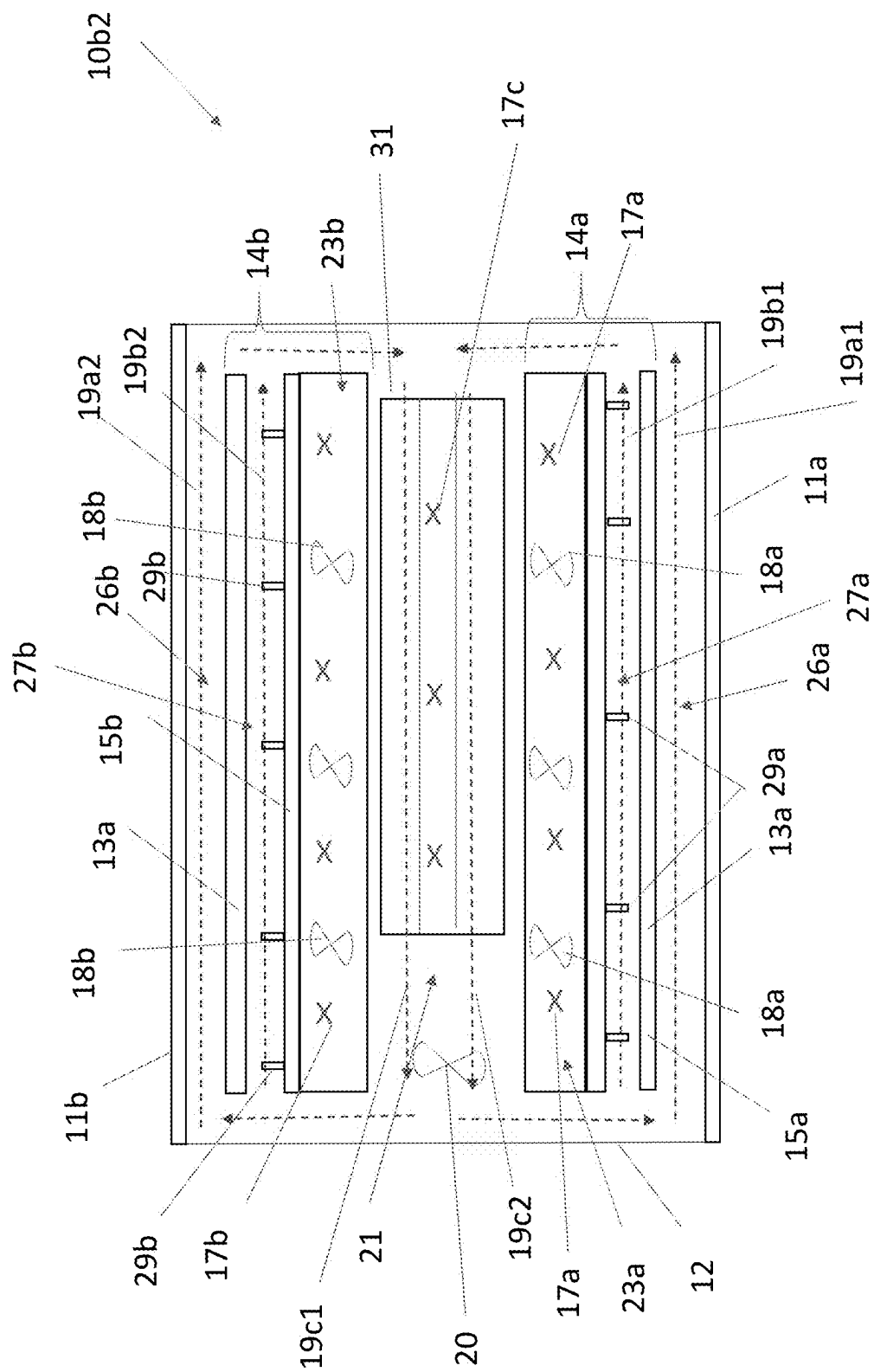


Figure 4B

DISPLAY ASSEMBLIES FOR PROVIDING COMPRESSIVE FORCES AT ELECTRONIC DISPLAY LAYERS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 18/920,434 filed Oct. 18, 2024, which is a continuation of U.S. application Ser. No. 18/609,896 filed Mar. 19, 2024, now U.S. Pat. No. 12,153,479 issued Nov. 26, 2024, which is a continuation of U.S. application Ser. No. 17/387,438 filed Jul. 28, 2021, now U.S. Pat. No. 11,966,263 issued Apr. 23, 2024, the disclosures of which are incorporated by reference as if fully restated herein.

TECHNICAL FIELD

Exemplary embodiments relate generally to display assemblies configured to provide compressive forces at on one or more electronic display layers, such as on one or both sides of electronic display layers.

BACKGROUND AND SUMMARY OF THE INVENTION

Certain electronic display layers, such as those comprising liquid crystals, are relatively thin. These electronic display layers often span a relatively large area. In some cases, these electronic display layers may only be supported at a perimeter. This may subject the electronic display layers to distortion by external forces, such as air pressure. Such distortions may result in various optical or other performance issues. These issues become particularly problematic when relatively large displays are utilized because relatively small air pressures experienced over a large area result in application of significant net forces. It is known to provide air on either side of an electronic display layer to create a pressure differential which reduces or eliminates bowing of the electronic display layer, such as is shown and/or described in U.S. Pat. No. 10,398,066 issued Aug. 27, 2019.

It is known to provide various optical layers at the electronics display layer and/or a cover for the same including, but not limited to, polarizing films, anti-reflective coatings, combinations thereof, or the like to improve optical qualities of displayed images. In certain cases, it may be desirable to provide compressive forces on one or both sides of an electronic display layer to prevent or eliminate cell breach of the electronic display layer and/or mechanical separation of optical layers. What is needed is a display assembly which provides compressive forces on one or both sides of an electronic display layer.

Display assemblies which provide compressive forces on one or both sides of an electronic display layer, and methods related to the same, are provided. The display assemblies may include one or more closed loop airflow pathways. At least one of the closed loop airflow pathways may include a front passageway which extends between a cover for the electronic display layer and the electronic display layer itself as well as an illumination device passageway which extends between the electronic display layer and an illumination device for the electronic display layer. A rear passageway extending behind the illumination device may be in fluid communication with both the front passageway and the illumination device passageway to complete the closed loop airflow pathway, which may encircle the illumination device and/or the electronic display layer.

A closed loop fan unit may be positioned adjacent to an entrance to the front passageway and the illumination device passageway in exemplary embodiments. The closed loop fan unit may be configured to push circulating gas through the front passageway and illumination device passageway when activated. The closed loop fan unit may be configured to generate a relatively high-pressure area at a first side of the closed loop fan unit facing the entrance to the front passageway and illumination device passageway such that such a relatively high-pressure flow is created within at least a portion of the front passageway and illumination device passageway when activated. The closed loop fan unit may be configured to generate a relatively low-pressure area at an opposing side of the closed loop fan unit facing the rear passageway when activated such that a relatively low-pressure flow is created within the rear passageway when activated. When operated, the closed loop fan unit may be configured to generate a pressure differential on one or both sides of the electronic display layer such that a pressure of a flow of the circulating gas in the front passageway is higher than a pressure of a flow of the circulating gas in the illumination device passageway. This may result in inward directed net forces at the electronic display layer which forces the electronic display layer rearward towards the illumination device, thereby reducing or eliminating bowing of the electronic display layer. This arrangement may be configured to create a positive pressure relative to ambient at one or both of a front and/or rear surface of the electronic display layer (e.g., in the front passageway and/or the illumination device passageway) so as to provide compressive forces at the front and/or rear side of the electronic display layer. This may prevent cell breach in the electronic display layer and/or mechanical separation of the optical layers from either or both of the electronic display layer and the cover. Optical spikes may be provided between the electronic display layer and the illumination device, such as within the illumination device passageway, to limit or prevent rearward movement of the electronic display layer towards the illumination device to prevent visual distortions.

In other exemplary embodiments, the closed loop fan unit may be positioned at an exit of the front passageway and illumination device passageway. The closed loop fan unit, in such embodiments, may be configured to draw circulating gas through the front passageway and the illumination device passageway. The closed loop fan unit, in such embodiments, may be configured to generate a relatively low-pressure area at the exit to the front passageway and the illumination device passageway and/or a relatively high-pressure area at an opposing side of the closed loop fan unit oriented towards the rear passageway when activated. In such embodiments, the closed loop fan unit may be configured to create a negative pressure relative to ambient within some or all of the front passageway and/or illumination device passageway such that a pressure of a flow of the circulating gas in the front passageway is higher than a pressure of a flow of the circulating gas in the illumination device passageway, even if one or both are negative relative to pressure of ambient air. Such an arrangement may permit creation of differential pressures sufficient to reduce or eliminate bowing of the electronic display layer.

The display assemblies may include multiple electronic display layers in exemplary embodiments. Where multiple electronic display layers are utilized, a single rear passageway may be common to at least two of the multiple electronic display layers. The display assemblies may include one or more open loop airflow pathways for ambient air. One or more open loop fan units may be provided for

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forcing ambient air through the open loop airflow pathways when operated. A structural framework may be used to secure and/or house the various components of the display assemblies. The closed loop fan units may be mounted within the rear passageways, in exemplary embodiments. Electronic components for operating the electronic display layers may be provided within the rear passageway.

Pressure sensors may be provided to monitor the differential pressure created, pressures of circulating gas within one or both of the front passageway and illumination device passageway and/or pressures of ambient air. Such sensors may be in electronic communication with one or more controllers. Operation of the closed loop fan units and/or open loop fan units may be adjusted, such as by the controllers, in response to readings from the sensors. For example, speed or duration of operation of the closed loop fan units may be adjusted to maintain the desired pressure differential and/or positive pressure relative to ambient within one or both of the front passageway and/or illumination device passageway.

Further features and advantages of the systems and methods disclosed herein, as well as the structure and operation of various aspects of the present disclosure, are described in detail below with reference to the accompanying figures.

BRIEF DESCRIPTION OF THE DRAWINGS

In addition to the features mentioned above, other aspects of the present invention will be readily apparent from the following descriptions of the drawings and exemplary embodiments, wherein like reference numerals across the several views refer to identical or equivalent features, and wherein:

FIG. 1 is a perspective view of an exemplary display assembly indicating section lines A-A;

FIG. 2A is a top sectional view of an exemplary embodiment of the display assembly of FIG. 1 taken along section line A-A;

FIG. 2B is a top sectional view of another exemplary embodiment of the display assembly of FIG. 1 taken along section line A-A;

FIG. 3A is a perspective sectional view of an exemplary embodiment of the display assembly of FIG. 1 taken along section line A-A;

FIG. 3A1 is a top sectional view of the display assembly of FIG. 3A;

FIG. 3A2 is a detailed perspective view of detail A of FIG. 3A1;

FIG. 3A3 is a detailed perspective view of detail C of FIG. 3A2;

FIG. 3A4 is a detailed top sectional view of detail A of FIG. 3A1;

FIG. 3A5 is a detailed top sectional view of detail B of FIG. 3A1;

FIG. 3B is a perspective sectional view of another exemplary embodiment of the display assembly of FIG. 1 taken along section line A-A;

FIG. 4A is a top sectional view of the display assembly of another exemplary embodiment of the display assembly of FIG. 1 taken along section line A-A; and

FIG. 4B is a top sectional view of the display assembly of another exemplary embodiment of the display assembly of FIG. 1 taken along section line A-A.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENT(S)

Various embodiments of the present invention will now be described in detail with reference to the accompanying

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drawings. In the following description, specific details such as detailed configuration and components are merely provided to assist the overall understanding of these embodiments of the present invention. Therefore, it should be apparent to those skilled in the art that various changes and modifications of the embodiments described herein can be made without departing from the scope and spirit of the present invention. In addition, descriptions of well-known functions and constructions are omitted for clarity and conciseness.

Embodiments of the invention are described herein with reference to illustrations of idealized embodiments (and intermediate structures) of the invention. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the invention should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

FIG. 1 is a perspective view of an exemplary electronic display assembly (hereinafter also a “unit”) 10 in accordance with the present invention. The unit 10 may include a structural framework 12. The structural framework 12 may be configured for mounting to a ground surface, such as a sidewalk or street, mounting to a wall or other surface, incorporation into street furniture (e.g., phone booths, bus shelters, benches, railings, combinations thereof, or the like), combinations thereof, or the like. The structural framework 12 may comprise one or more members, panels, cladding, panels, housings, combinations thereof, or the like.

The units 10 may comprise one or more electronic display subassemblies 14. Some or all of the electronic display subassemblies 14 may be attached to the structural framework 12 in a moveable manner, though such is not required. For example, the electronic display subassemblies 14 may be attached to the structural framework in a hinged manner to permit selective movement between a closed position whereby certain parts of the units 10 are fully or partially sealed, and an open position whereby certain parts of the interior of the unit 10 are exposed for access.

One or more intakes and exhausts 16 may be provided at the units 10 for ingesting and exhausting ambient air.

FIG. 2A through FIG. 4B illustrate various exemplary embodiments of airflow pathways within the units 10. FIGS. 2A-3B illustrate exemplary units 10a1, 10a2 with a single electronic display subassembly 14. FIGS. 4A-4B illustrate exemplary units 10b1, 10b2 with two electronic display subassemblies 14a, 14b placed in a back-to-back arrangement. Any number of electronic display subassemblies 14 may be utilized in any arrangement with the structural framework 12. Similar or the same components used in conjunction with units 10 having multiple electronic display subassemblies 14 may use the same numbering with the addition of an “a”, “b” and/or “1”, “2”, etc. (e.g., 14 to 14a, 14b, 19a to 19a1, 19a2). Any features or arrangements shown and/or described with respect to any of the embodiments of the units 10a1, 10a2, 10b1, 10b2 (sometimes referred to collectively as “unit 10” or “units 10” herein) may be used with any other embodiment of the units 10.

Each electronic display subassembly 14 may comprise an illumination device 15. In exemplary embodiments, the illumination device 15 may comprise a number of lighting elements, such as LEDs, provided at a substrate, such as a printed circuit board. Each electronic display subassembly 14 may comprise an electronic display layer 13. The electronic display layer 13 may comprise a layer of liquid crystals, such as for a liquid crystal display, though any type

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or kind of electronic display may be utilized. In exemplary embodiments, the illumination device **15** may be provided rearward of the electronic display layer **13** to serve as a direct backlight. In other exemplary embodiments, the illumination device **15** may comprise one or more diffusive and/or transmissive layers and the substrate and/or lighting elements may be positioned about the edge of the electronic display layer **13** to provide edge lighting to the same.

The electronic display layer **13** and/or illumination device **15** may be positioned rearward of a cover **11**. The cover **11** may comprise one or more layers of a transparent or translucent material(s). In exemplary embodiments, each cover **11** may comprise two layers bonded with an optically clear adhesive, which may provide increased impact protection. One or more polarizers, anti-reflective materials, combinations thereof, or the like may be disposed on some or all of the cover **11** as a coating, film, layer, combinations thereof, or the like. The cover **11** may form part of the electronic display subassembly **14** or may be separate therefrom. The cover **11** and the structural framework **12** may together substantially enclose the units **10**, such as with intakes/exhausts **16** exempted, when the subassemblies **14** are placed in a closed position. The cover **11** may be configured to move with the electronic display subassembly **14**, may be configured for independent movement, and/or may be fixed to the structural framework **12**. Each of the electronic display subassemblies **14** may be connected to the structural framework **12** in a hinged or otherwise movable manner, though such is not required.

A single or multiple such electronic display subassemblies **14** may be provided at a single unit **10**, such as, but not limited to, in a back-to-back arrangement. The electronic display subassemblies **14** may be of the same or different type and may comprise the same or different components. The electronic display subassemblies **14** may be provided in any arrangement such as portrait or landscape.

The intakes and/or exhausts **16** may be fluidly connected to one or more open loop airflow pathways **23** within the units **10**. A respective one of the open loop airflow pathways **23a**, **23b** may extend through a respective one of the electronic display subassemblies **14a**, **14b** in exemplary embodiments such that an open loop airflow pathway **34** is provided for each one of the electronic display subassemblies **14**, which may be entirely separate, or separated for a distance and rejoined. For example, without limitation, the open loop airflow pathways **23** may extend behind and along at least a portion of the illumination devices **15** for the electronic display layer **13** for accepting flows **17a**, **17b** of ambient air. The open loop airflow pathways **23** may comprise one or more corrugated layers **25** in exemplary embodiments.

One or more closed loop airflow pathways may be provided within the units **10**. In exemplary embodiments, such closed loop airflow pathways may comprise at least a front passageway **26**, which may extend between the cover **11** and the electronic display layer **13**. Such closed loop airflow pathways may comprise a rear passageway **21**, which may extend behind the electronic display subassembly **14**, and/or behind the electronic display layer **13**. Such closed loop airflow pathways may extend entirely within the unit **10**, such as within outer boundaries of the structural framework **12**. Where multiple electronic display subassemblies **14a**, **14b** are utilized, the rear passageway **21** may be common to at least two of the electronic display subassemblies **14a**, **14b**, though such is not required. A heat exchanger **31** may be located within the rear passageway **21**, though such is not required. The heat exchanger **31** may comprise a multilayer

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heat exchanger configured to accommodate a common flow **19c** of the circulating gas through at least some of the layers as well as one or more flows **17c** of ambient air through at least some other ones of the layers.

Various electronic components **35** for operating the unit **10** may be provided within the rear passageway **21**. The electronic components **35** may include, for example, without limitation, video players, power supplies, processors, electronic storage devices, controllers **37**, sensors **44**, combinations thereof, or the like. Any number, type, and/or kind of components **35** may be utilized.

An illumination device passageway **27a**, **27b** may extend between each of the electronic display layers **13a**, **13b** and the respective illumination devices **15a**, **15b**. A flow of circulating gas **19c** within the rear passageway **21** may be separated such that a first portion **19a** flows through the front passageway **26** and a second portion **19b** flows through the illumination device passageway **27**. The flows **19a**, **19b** may be recombined, such as in the rear passageway **21**.

One or more open loop fan units **18** may be provided. The same of different open loop fan units **18** may be associated with each of the open loop airflow pathways **23**. The open loop fan units **18** may be configured to ingest ambient air **17** into the units **10**, exhaust ambient air **17** from the assembly **10**, and/or move ingested ambient air **17** through the one or more open loop airflow pathways **23** when activated. Multiple open loop fan units **18a**, **18b** may be used where multiple electronic display subassemblies **14a**, **14b** are utilized, for example, or such open loop fan units **18** may be common to multiple ones of the electronic display subassemblies **14** of such units **10**.

One or more closed loop fan units **20** may be provided. Each closed loop fan unit **20** may comprise one or more fans or the same of different type. The same or different closed loop fan units **20** may be associated with each of the closed loop airflow pathways. The closed loop fan units **20** may be configured to move circulating gas through said one or more closed loop airflow pathways when activated. The closed and open loop fan units **20**, **18** may comprise axial fans, centrifugal fans, combinations thereof, or the like. Any number or type of fan units **20**, **18** may be used at any location in the units **10**, and may be provided in banks or sets. The open loop airflow pathways **23** may be separate from the closed loop airflow pathways, though a complete (e.g., gas impermeable) separation is not necessarily required. The same closed loop fan units **20** may be used to move circulating gas through the front passageways **26a**, **26b** and/or illumination device passageways **27a**, **27b** of multiple electronic display subassemblies **14a**, **14b** where such multiple electronic display subassemblies **14a**, **14b** are provided for a unit **10**, though such is not required.

The unit **10** may comprise a controller **37** and/or one or more sensors **44**. The sensors **44** may comprise, for example, without limitation, temperature sensors, fan speed sensors, airflow sensors, humidity sensors, relative humidity sensors, air pressure sensors, differential pressure sensors, location sensors, moisture sensors, combinations thereof, or the like. Any type, kind, or number of sensors **44** may be utilized at any number of locations within the units **10**. In exemplary embodiments, at least certain of the sensors **44** may comprise tubes or other fluid passageways to connect the sensor (s) **44** to other parts of the units **10** and/or the ambient environment. This may provide for flexibility in placement and design. Any type, kind, or number of controllers **31** may be utilized at any number of locations within the units **10**. Such sensors **44** and/or controller **37** are not necessarily required, and may be omitted from view of certain embodi-

ments provided herein to more clearly illustrate other components. However, such sensors **44** and/or controller **37** may be utilized in the same or similar arrangements in such embodiments, even where not expressly provided in the figures.

The one or more closed loop fan units **20** may be configured to create a pressure differential. For example, an intake side of each closed loop fan unit **20** may be configured to generate a relatively low-pressure area or flow and an exhaust side may be configured to generate a relatively high-pressure area or flow. As illustrated with particular regard to FIGS. **2A**, **3A-3A5**, and **4A**, in exemplary embodiments, one or more of the closed loop fan units **20** may be positioned adjacent to an exhaust for the front passageway **26** and/or the illumination device passageway **27**. For example, without limitation, the closed loop fan units **20** in such embodiments may be positioned at a portion of the rear passageway **21** adjacent to an exit from one or both of the front passageway **26** and the illumination device passageway **27** for generating the flows **19a** and/or **19b** within the front passageway **26** and the illumination device passageway **27** by pulling the circulating gas into an intake side of the closed loop fan unit **20**, and generating the flow **19c** within the rear passageway **21** by pushing the circulating gas out an exhaust side of the closed loop fan unit **20**. Because the intake, relatively low-pressure side, of the closed loop fan units **20** is fluidly adjacent to the front passageway **26** and/or the illumination device passageway **27**, the pressure of the flows **19a** and/or **19b**, particularly near the exit of the front passageway **26** and/or the illumination device passageway **27**, may be maintained at a relatively low level, such as negative relative to pressure of ambient air outside of the units **10**, though such is not necessarily required.

Alternatively, without limitation, the one or more of the closed loop fan units **20** may be positioned at a portion of the rear passageway **21** adjacent to an entrance into one or both of the front passageway **26** and the illumination device passageway **27** for generating the flows **19a** and/or **19b** through the front passageway **26** and the illumination device passageway **27** by pushing the circulating gas through the front passageway **26** and the illumination device passageway **27**. Because the exhaust, relatively high-pressure side of the closed loop fan units **20** is fluidly adjacent to the front passageway **26** and/or the illumination device passageway **27**, the pressure of the flows **19a** and/or **19b**, particularly at the entrances to the front passageway **26** and the illumination device passageway **27**, may be maintained at a relatively high level, such as greater than pressure of ambient air outside of the units **10**, though such is not necessarily required.

The front passageway **26** and/or the rear passageway **21** may be configured to create and maintain a pressure differential between the flows **19a** and **19b** of the circulating gas in the front passageway **26** and/or the rear passageway **21** sufficient to generate net forces at the electronic display layers **13** which reduces or eliminates bowing of the electronic display layers **13**. In exemplary embodiments, the pressure of the flow **19a** in the front passageway **26** may be maintained at a higher level than the flow **19b** in the illumination device passageway **27**, resulting in rearward forces against the electronic display layer **13** to reduce or eliminate outward bowing. Such pressure differentials may be generated using features including, but not limited to, those shown and/or described in U.S. Pat. No. 10,398,066 issued Aug. 27, 2019, the disclosures of which are hereby incorporated by reference as if fully restated herein.

In exemplary embodiments, without limitation, one or more intake openings **53a**, **53b** into the illumination device passageway **27** from the rear passageway **21** may be larger than one or more exit openings **55a**, **55b** from the illumination device passageway **27** to the rear passageway **21**. In this manner, more circulating gas may be ingested into the illumination device passageway **27** than is readily able to exit, thereby raising average pressure in the illumination device passageway **27**.

As shown with particular regard to FIGS. **3A1-3A5**, the electronic display layer **13** may be held in place by one or more brackets **51a**, **51b**. The brackets **51** may define, at least in part, the one or more intake openings **53a**, **53b** and/or the one or more exit openings **55a**, **55b**. At least some of the brackets **51** may comprise protrusions **57** configured to maintain relatively spacing for the intake openings **53** and/or the exit openings **55**.

One or more of the closed loop fan units **20** may be provided wholly or partially within one or more housings **59**. The housings **59** may be configured to direct circulating gas in an appropriate direction, such as between the front passageway **26** and/or the illumination device passageway **27** into the rear passageway **21**, or vice versa.

One or more gaskets **61** may be provided at the housing **59**. For example, without limitation, the gaskets **61** may be provided to fully or partially seal the housings **59** against a panel **63**. In this fashion, the closed loop fan units **20** may be fully or partially sealed against the panel **63** so as to increase the amount of air forced into the front passageway **26** and/or the illumination device passageway **27**, such as from the rear passageway **21**, and/or out of the front passageway **26** and/or the illumination device passageway **27**, such as from the rear passageway **21**. The gaskets **61** may, alternatively or additionally, provide vibrational damping and/or sound reduction, such as during operation of the closed loop fan units **20**. The gaskets **61** may comprise rubber and/or a rubber composite, by way of non-limiting example.

The panel **63** may form part of the open loop airflow pathway **23**. For example, without limitation, the panel **63** and an additional panel **65** may be provided spaced apart from one another, such as with the corrugated layer **25** therebetween, to at least partially define the open loop airflow pathway **23**.

In exemplary embodiments, positive pressure may be maintained in only the front passageway **26** and/or relatively high pressure may be maintained in the front passageway **26** (e.g., relative to pressure in the flow **19b** in the illumination device passageway **27**) such that the electronic display layer **13** is pushed towards the illumination device **15** when the closed loop fan unit **20** is operated. Optical spikes **29** or other support structures may be utilized within the illumination device passageway **27** to reduce or eliminate movement of the electronic display layer **13** towards the illumination device **15**, such as past the optical spikes **29**. The optical spikes **29** may comprise one or more optically transmissible materials. The optical spikes **29** may comprise rods, cones, or the like positioned within the illumination device cavity **27** and may be configured to limit or prevent rearward travel of the electronic display layer **13**. The optical spikes **29** may exert normal, compressive forces on the electronic display layer **13**, particularly in conjunction with the pressure of the flow **19a** of the circulating gas within the front passageway **26**. This may be particularly beneficial when unable to generate positive or sufficiently high pressure for the flow **19b** of the circulating gas within the illumination device passageway **27**. This may occur, for

example, without limitation, due to variations in ambient air and/or circulating gas pressure. Circulating gas pressure, in particular, may vary due to temperature variations in the circulating gas (e.g., due to solar loading) and/or ambient temperatures, which may affect the unit's 10 ability to remove heat in air-to-air heat exchange.

In exemplary embodiments, the pressure of the circulating gas in the front passageway 26 and the illumination device passageway 27 may be regularly, sporadically, and/or continuously monitored and operations may be adjusted based on such measurements. Such operations may include the speed or other operation of the closed loop fans 20. Such operations may be adjusted to desired pressures in the front passageway 26 and the illumination device passageway 27, such as positive pressures in one or both, and/or desired differential pressure between the same.

Any number, type, kind, and/or arrangement of such optical spikes 29 may be utilized. In embodiments where more than one electronic display layer 13a, 13b is utilized, more than one set of optical spikes 29a, 29b for each respective one of the electronic display layers 13a, 13b of the same or different type may likewise be utilized, though such is not required. Such optical spikes 29 are not necessarily required, and may be omitted from view of certain embodiments provided herein to more clearly illustrate other components. However, such optical spikes 29 may be utilized in the same or similar arrangements in such embodiments, even if not expressly provided in the figures.

Any embodiment of the present invention may include any of the features of the other embodiments of the present invention. The exemplary embodiments herein disclosed are not intended to be exhaustive or to unnecessarily limit the scope of the invention. The exemplary embodiments were chosen and described in order to explain the principles of the present invention so that others skilled in the art may practice the invention. Having shown and described exemplary embodiments of the present invention, those skilled in the art will realize that many variations and modifications may be made to the described invention. Many of those variations and modifications will provide the same result and fall within the spirit of the claimed invention. It is the intention, therefore, to limit the invention only as indicated by the scope of the claims.

Certain operations described herein may be performed by one or more electronic devices. Each electronic device may comprise one or more processors, electronic storage devices, executable software instructions, and the like configured to perform the operations described herein. The electronic devices may be general purpose computers or specialized computing devices. The electronic devices may comprise personal computers, smartphone, tablets, databases, servers, or the like. The electronic connections and transmissions described herein may be accomplished by wired or wireless means. The computerized hardware, software, components, systems, steps, methods, and/or processes described herein may serve to improve the speed of the computerized hardware, software, systems, steps, methods, and/or processes described herein.

What is claimed is:

1. A display assembly for inhibiting delamination of an optical stack, said display assembly comprising:
 - a structural framework defining, at least in part, an enclosure, wherein the optical stack is positioned within the enclosure and includes an electronic display layer;
 - one or more airflow passageways located within the enclosure and defining, at least in part, a continuous,

interconnected airflow pathway extending, at least in part, along one or more surfaces of the optical stack; and

one or more fan units positioned along the continuous, interconnected airflow pathway, which when activated, move gas through the continuous, interconnected airflow pathway and generate compressive forces at the optical stack.

2. The display assembly of claim 1 wherein:

the one or more airflow passageways comprise a first airflow passageway located, at least in part, forward of the optical stack; and

said one or more fan units, when activated, at least primarily push said gas through said first airflow passageway.

3. The display assembly of claim 2 wherein:

at least one of the one or more fan units is positioned at or proximate to an entrance to the first airflow passageway.

4. The display assembly of claim 2 wherein:

the one or more airflow passageways comprise a second airflow passageway located, at least in part, rearward of a forwardmost layer of the optical stack.

5. The display assembly of claim 4 wherein:

said one or more fan units, when activated, at least primarily push said gas through said first airflow passageway and said second airflow passageway.

6. The display assembly of claim 5 wherein:

at least one of the one or more fan units is positioned at or proximate to an entrance to each of the first airflow passageway and said second airflow passageway.

7. The display assembly of claim 4 wherein:

the one or more airflow passageways comprise a third airflow passageway located, at least in part, rearward of the optical stack;

said one or more fan units, when activated, at least primarily push said gas through said second airflow passageway; and

said one or more fan units, when activated, at least primarily pulls said gas through said third airflow passageway.

8. The display assembly of claim 7 wherein:

at least one of the one or more fan units is positioned at or proximate to an entrance to each of the first airflow passageway and said second airflow passageway and at or proximate to an exit of the third airflow passageway; and

said one or more fan units, when activated, pushes said gas at a relatively high-pressure into the entrance to said first and second passageways and pulls said gas at a relatively low-pressure side from the exit of said third passageway.

9. The display assembly of claim 1 wherein:

said one or more airflow passageways comprise a first airflow passageway positioned forward of the optical stack, a second airflow passageway positioned rearward of a forwardmost layer of the optical stack, and a third airflow passageway located rearward of the optical stack;

at least a majority of the first, second, and third airflow passageways extend in longitudinal alignment with one another; and

said one or more fan units, when activated, moves said gas through said first and second airflow passageways in a first, same primary direction and through said third airflow passageway in a second primary direction opposing the first direction.

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10. The display assembly of claim 1 wherein:
each of said one or more airflow passageways comprise at
least one duct.
11. The display assembly of claim 1:
wherein the electronic display layer comprises liquid 5
crystals;
wherein the optical stack comprises at least one of: a
polarizer, an anti-reflective film, an optical enhance-
ment film, a diffusive layer, and an optically transmis-
sive layer; and 10
further comprising an illumination device arranged to
provide direct backlighting to the electronic display
layer when operated.
12. The display assembly of claim 11 wherein:
an open airflow pathway for ambient air; and 15
a second set of one or more fan units, when activated,
moves said ambient air through said open airflow
pathway, wherein said continuous, interconnected air-
flow pathway is partitioned from an ambient environ-
ment and the open airflow pathway.
13. The display assembly of claim 12 wherein:
said open airflow pathway extends, at least in part, along
a rear surface of the illumination device.
14. The display assembly of claim 1 further comprising:
a cover provided forward of, and spaced apart from, said 25
optical stack to define, at least in part, the first airflow
passageway and the enclosure for the optical stack.
15. The display assembly of claim 1 wherein:
said optical stack is part of an electronic display subas-
sembly which is moveably connected to the structural 30
framework.
16. The display assembly of claim 1 further comprising:
a differential pressure sensor fluidly connected to, and
configured to measure a pressure of, ambient air and
said gas in at least a portion of the continuous, inter- 35
connected airflow pathway; and
- a controller in electronic communication with said differ-
ential pressure sensor and said one or more fans unit,
said controller comprising software instructions, which 40
when executed, configure the controller to adjust opera-
tions of at least one of said one or more fan units in
response to measurements from said differential pres-
sure sensor to maintain a positive difference between
the measured pressure of the gas in at least the portion 45
of the continuous, interconnected airflow pathway and
the measured pressure of the ambient air.
17. The display assembly of claim 1 further comprising:
a second optical stack having a front surface facing an
opposing direction from a front surface of said optical
stack, wherein said continuous, interconnected airflow 50
pathway extends along one or more surfaces of the
second optical stack.
18. The display assembly of claim 17 wherein:
said one or more airflow passageways comprise a first
airflow passageway positioned forward of the front 55
surface of the optical stack, a second airflow passage-

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- way positioned forward of the front surface of the
second optical stack, and a third airflow passageway
located between the optical stack and the second optical
stack;
at least a majority of the first, second, and third airflow
passageways extend in longitudinal alignment with one
another; and
said one or more fan units, when activated, moves said gas
through said first and second airflow passageways in a
first, same primary direction and through said third
airflow passageway in a second primary direction
opposing the first direction.
19. A method for inhibiting delamination of the optical
stack of the display assembly of claim 1, said method
comprising:
operating the one or more fan units.
20. A display assembly for inhibiting delamination of
optical stacks, said display assembly comprising:
a structural framework defining, at least in part, an
enclosure, wherein a first one of the optical stacks and
a second one of the optical stacks is positioned within
the enclosure, and wherein each of said optical stacks
includes an electronic display layer comprising liquid
crystals and a direct backlight for the respective elec-
tronic display layer;
one or more airflow passageways located within the
enclosure and defining a continuous, interconnected
airflow pathway extending, at least in part, along one or
more surfaces of the optical stack, said one or more
airflow passageways comprise a first airflow passage-
way positioned forward of a front surface of the first
one of the optical stacks, a second airflow passageway
positioned forward of a front surface of the second one
of the optical stacks, and a third airflow passageway
located between the first one of the optical stacks and
the second one of the optical stacks;
cover layers, wherein a first one of said cover layers is
positioned forward of the front surface of the first one
of the optical stacks and defines, at least in part, said
first airflow passageway and said enclosure, and
wherein a second one of said cover layers is positioned
forward of the front surface of the second one of the
optical stacks and defines, at least in part, said second
airflow passageway and said enclosure; and
one or more fan units positioned along the continuous,
interconnected airflow pathway at or proximate to an
entrance to the first airflow passageway and the second
airflow passageway and at or proximate to an exit of the
third airflow passageway such that the one or more fan
units, when activated, move gas through the continu-
ous, interconnected airflow pathway and generate com-
pressive forces at each of the optical stacks.

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